

Chapter One

Introduction to Artificial Intelligence

Outlines

- ▶ Introduction to AI
 - ▶ Objectives/Goals of AI
 - ▶ What is AI?
 - ▶ Approaches to AI – making computer:
 - ▶ Think like a human (Thinking humanly)
 - ▶ Act like a human (Acting humanly)
 - ▶ Think rationally (Thinking rationally)
 - ▶ Act rationally (Acting rationally)
 - ▶ The Foundations of AI
- ▶ Bits of History and the State of the Art

Introduction

- ▶ We call ourselves Homo sapiens—**man the wise**—because our **intelligence** is so important to us.
- ▶ For thousands of years, we have tried to understand how **we think**; that is, how a mere handful of matter can **perceive, understand, predict, and manipulate** a world far larger and more complicated than itself.
- ▶ The field of **artificial intelligence**, or AI, goes further still: it attempts not just to understand but also to **build intelligent entities**.

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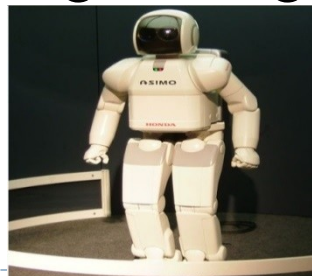
- ▶ AI currently encompasses a huge variety of subfields, ranging from the general (**learning and perception**) to the specific, such as playing chess, proving mathematical theorems, writing poetry, driving a car on a crowded street, and diagnosing diseases.
- ▶ AI is relevant to any intellectual task; it is truly a universal field.

Basic Concepts

- ▶ There are eight definitions of AI, laid out along two dimensions.
- ▶ The definitions on top are concerned with **thought processes and reasoning**, whereas the ones on the bottom address **behavior**.
- ▶ The definitions on the left measure success in terms of fidelity to **human performance**, whereas the ones on the right measure against an **ideal performance** measure, called **rationality**.
- ▶ A system is rational if it does the “**right thing**,” given **what it knows**

What is AI?

- ▶ **Artificial Intelligence (AI)** is usually defined as the science of making computers do things that require **intelligence** when done by humans.
- ▶ AI has had some success in limited, or simplified, domains.
- ▶ However, the five decades since the inception of AI have brought only very slow progress, and early optimism concerning the attainment of human-level intelligence has given way to an appreciation of the profound difficulty of the problem.
- ▶ The term AI is first used by John McCarthy (1956) who considers it to mean the science and engineering of making **intelligent machine**.



What is Intelligence

- ▶ **Intelligence** is a general mental capability that involves the ability to reason, plan, solve problems, think abstractly, comprehend ideas and language, and learn.
- ▶ Quite simple human behavior can be intelligent yet quite complex behavior performed by insects is unintelligent.
- ▶ What is the difference? Consider the behavior of the digger wasp, *Sphex ichneumoneus*.
- ▶ When the female wasp brings food to her burrow, she deposits it on the threshold, goes inside the burrow to check for intruders, and then if the coast is clear carries in the food.

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- ▶ The unintelligent nature of the *wasp's* behavior is revealed if the watching experimenter moves the food a few inches while the wasp is inside the burrow checking. On emerging, the wasp repeats the whole procedure: she carries the food to the threshold once again, goes in to look around, and emerges. She can be made to repeat this cycle of behavior upwards of forty times in succession.
 - ▶ Intelligence--conspicuously absent in the case of *Sphex*--is the ability to adapt one's behavior to fit new circumstances.
 - ▶ Mainstream thinking in psychology regards human intelligence not as a single ability or cognitive process but rather as an array of separate components.
 - ▶ Research in AI has focused chiefly on the following components of intelligence: **learning, reasoning, problem-solving, perception, and language-understanding.**
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Learning

- ▶ Learning is distinguished into a number of different forms.
- ▶ The simplest is learning by **trial-and-error**.
- ▶ For example, a simple program for solving **mate-in-one** chess problems might try out moves at random until one is found that achieves mate.
- ▶ The program remembers the successful move and next time the computer is given the same problem it is able to produce the answer immediately. The simple memorizing of individual items--solutions to problems, words of vocabulary, etc.--is known as **rote learning**.
- ▶ Rote learning is relatively easy to implement on a computer. More challenging is the problem of implementing what is called **generalization**.

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- ▶ Learning that involves **generalization** leaves the learner able to perform better in situations not previously encountered.
 - ▶ *A program that learns past tenses of regular English verbs by **rote** will not be able to produce the past tense of e.g. "jump" until presented at least once with "jumped", whereas a program that is able to **generalize** from examples can learn the "add-ed" rule, and so form the past tense of "**jump**" in the absence of any previous encounter with this verb.*
 - ▶ **Sophisticated modern techniques enable programs to generalize complex rules from data.**

Reasoning

- ▶ To reason is to **draw inferences** appropriate to the situation in hand. Inferences are classified as either **deductive** or **inductive**.
- ▶ **Deductive**: "Fred is either in the museum or the cafe; he isn't in the cafe; so he's in the museum",
- ▶ **Inductive**: "Previous accidents just like this one have been caused by instrument failure; so probably this one was caused by instrument failure".
- ▶ The difference between the two is that in the deductive case, the truth of the premises guarantees the truth of the conclusion, whereas in the inductive case, the truth of the premises lends support to the conclusion that the accident was caused by instrument failure, but nevertheless further investigation might reveal that, despite the truth of the premises, the conclusion is in fact false.
- ▶ There has been considerable success in programming computers to draw inferences, especially deductive inferences.
- ▶ Reasoning involves drawing inferences that are relevant to the task or situation in hand.

Problem-solving

- ▶ Problems have the general form: given such-and-such data, find x . A huge variety of types of problem is addressed in AI.
- ▶ Some examples are: finding winning moves in board games; identifying people from their photographs; and planning series of movements that enable a robot to carry out a given task.
- ▶ Problem-solving methods divide into **special-purpose** and **general-purpose**.
- ▶ A **special-purpose** method is tailor-made for a particular problem, and often exploits very specific features of the situation in which the problem is embedded.
- ▶ A **general-purpose** method is applicable to a wide range of different problems.
- ▶ One **general-purpose** technique used in AI is means-end analysis, which involves the step-by-step reduction of the difference between the current state and the goal state.
- ▶ The program selects actions from a list of means--which in the case of, say, a simple robot, might consist of **pickup**, **putdown**, **moveforward**, **moveback**, **moveleft**, and **moveright**--until the current state is transformed into the goal state.

Perception

- ▶ In perception the environment is scanned by means of various sense-organs, real or artificial, and processes internal to the perceiver analyze the scene into objects and their features and relationships.
- ▶ Analysis is complicated by the fact that one and the same object may present many different appearances on different occasions, depending on the angle from which it is viewed, whether or not parts of it are projecting shadows, and so forth.
- ▶ At present, artificial perception is sufficiently well advanced to enable a self-controlled car-like device to drive at moderate speeds on the open road, and a mobile robot to roam through a suite of busy offices searching for and clearing away empty soda cans.

Language-understanding

- ▶ A language is a system of signs having meaning by convention. Traffic signs, for example, form a mini-language, it being a matter of convention that, for example, the hazard-ahead sign means hazard ahead.
 - ▶ This meaning-by-convention that is distinctive of language is very different from what is called natural meaning, exemplified in statements like 'Those clouds mean rain' and 'The fall in pressure means the valve is malfunctioning'.
 - ▶ An important characteristic of full-fledged human languages, such as English, which distinguishes them from, e.g. bird calls and systems of traffic signs, is their *productivity*.
 - ▶ A productive language is one that is rich enough to enable an unlimited number of different sentences to be formulated within it.
 - ▶ It is relatively easy to write computer programs that are able, in severely restricted contexts, to respond in English, seemingly fluently, to questions and statements, for example the [Parry](#) and [Shrdlu](#) programs described in the section Early AI Programs. However, neither Parry nor Shrdlu actually understands language.
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Introduction ... Cont'd

- ▶ AI system possess
 - ▶ **Natural language processing** to enable it to communicate successfully in human language
 - ▶ **Knowledge representation** to store what it knows or hears
 - ▶ **Automated reasoning** to use the stored information to answer questions and to draw new conclusions
 - ▶ **Machine learning** to adapt to new circumstances and to detect and extrapolate patterns
 - ▶ **Computer vision** to perceive objects
 - ▶ **Robotics** to manipulate objects and move about
 - ▶ And others

Some Application Areas of AI

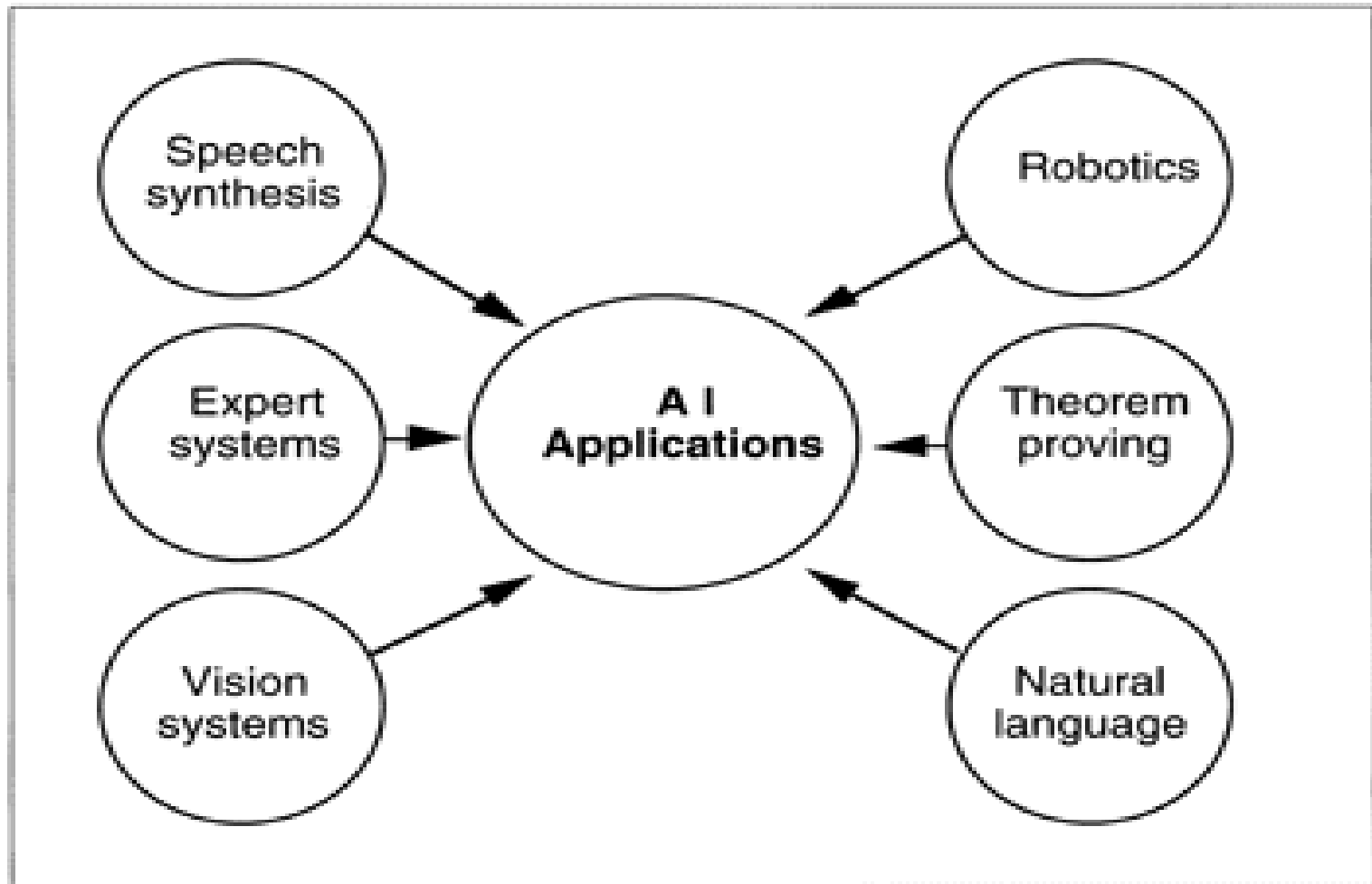


Figure 1. Some applications of Artificial Intelligence

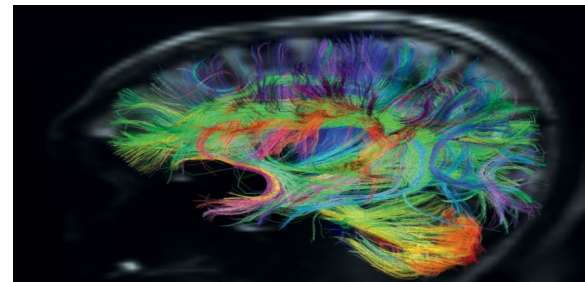
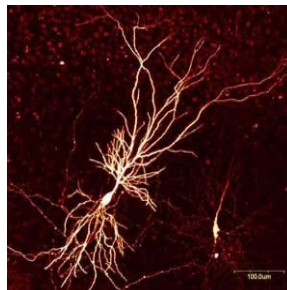
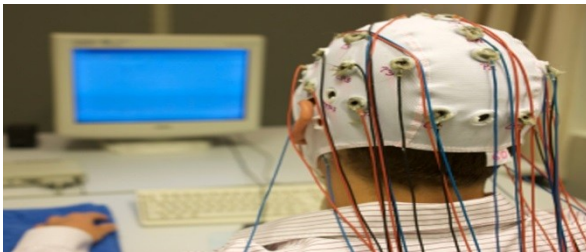
Views of defining AI

- ▶ Different scholars define AI differently

1. Thinking humanly	2. Acting humanly
3. Thinking rationally	4. Acting rationally

AI definition 1: Thinking humanly

- Need to study the brain as an information processing machine: cognitive science and neuroscience
- ▶ **AI as systems that think humanly**
 - ▶ “The automation of activities that we associate with human thinking, activities such as decision-making, problem solving, learning ...” (**Bellman** definition, 1978)
 - ▶ “The exciting new effort to make computers think ... machines with minds, in the full and literal sense” (**Haugeland** definition, 1985)



Thinking humanly: cognitive modeling

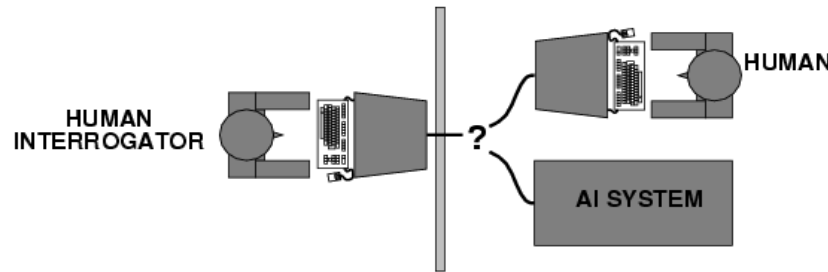
- Can we build a brain?
- ▶ **Requires:**
 - ▶ Scientific theories of internal activities of the brain
 - ▶ How human thinks?
- ▶ **How to validate a given agent think humanly?**
- ▶ The answer Requires either
 1. Predicting and testing the behavior of human subjects from his/her thinking point of view (**top-down**) or
 2. Direct identification from **neurological** data (**bottom-up**)
- ▶ Study on Mental processing logic of human being (cognitive science) is not yet fertile

AI definition 2: Acting humanly

- ▶ “The art of creating machines that perform functions that require intelligence when performed by people.” (Kurzweil definition, 1990)
- ▶ “The study of how to make computers do things at which, at the moment, people are better.” (Rich and Knight definition, 1991)

Acting humanly: Turing Test

- ❖ Turing (1950) on his famous paper "Computing machinery and intelligence":
- ❖ "Can machines think?" → "Can machines behave intelligently?"
- ❖ Operational test for intelligent behavior: the Imitation Game

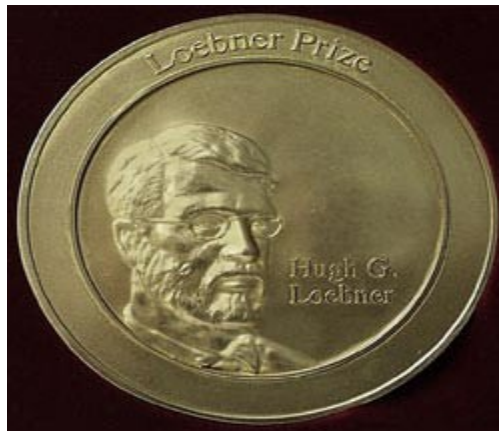


- ❖ Predicted that by 2000, a machine might have a 30% chance of fooling a person for 5 minutes
- ❖ Anticipated all major arguments against AI in following 50 years
- ❖ Active areas of research to achieve this: Machine learning, NLP, Computer vision, etc

The Turing Test

- **Loebner prize**

- ▶ 2008 competition: each of 12 judges was given five minutes to conduct simultaneous, split-screen conversations with two hidden entities (human and chatterbot). The winner, Elbot of Artificial Solutions, managed to fool three of the judges into believing it was human [**Wikipedia**].



AI definition 3: Thinking rationally

- ▶ A system is said to be rational if it does the “right thing” given what it knows.
- ▶ “The study of mental faculties through the use of computational models.” (Charniak and McDermott definition, 1985)
- ▶ “The study of the computations that make it possible to perceive, reason, and act.” (Winston definition, 1992)

Thinking rationally: "laws of thought"

- ▶ Right thinking is related to irrefutable reasoning process
- ▶ Require structure that always gave correct conclusion given correct premises
- ▶ Logic is the key to design and implement an agent that think rationally
- ▶ Several Greek schools developed various forms of *logic: notation* and *rules of derivation* for thoughts;
- ▶ Direct line through mathematics and philosophy to modern AI
- **Logic:** patterns of argument that always yield correct conclusions when supplied with correct premises
 - ▶ **“Socrates is a man; all men are mortal; therefore Socrates is mortal.”**

AI definition 3: Thinking rationally

- **Logicist approach to AI:** describe problem in formal logical notation and apply general deduction procedures to solve it
- Problems with the logicist approach
 - ▶ Computational complexity of finding the solution
 - ▶ Describing real-world problems and knowledge in logical notation
 - ▶ Dealing with uncertainty
 - ▶ A lot of “rational” behavior has nothing to do with logic

AI definition 4: Acting rationally

- ▶ “Computational intelligence is the study of the design of intelligent agent” (Poole, et al definition, 1998)
- ▶ “AI ... is concerned with intelligent behavior in artifacts.” (Nilsson definition, 1998)
- ▶ The course advocates to agents that act rationally

Acting rationally: rational agent

- ▶ Means acting so as to achieve one's goals, given one's beliefs.
- ▶ In this approach, AI is viewed as the study and construction of rational agent
- ▶ **Rational behavior:** doing the right thing
- ▶ **The right thing:** is the action/decision which is expected to **maximize** goal achievement, given the available information
- ▶ Doesn't necessarily involve thinking
 - ▶ One way of acting rationally is to reason logically to the action.
 - ▶ This indicates, making correct inference is part of being a rational agent
 - ▶ But rationality doesn't require correct inference because some time without having correct thing to do, agent must act rationally

Utility maximization formulation

- Advantages
 - ▶ Generality: goes beyond explicit reasoning, and even human cognition altogether
 - ▶ Practicality: can be adapted to many real-world problems
 - ▶ Naturally accommodates uncertainty
 - ▶ Amenable to good scientific and engineering methodology
 - ▶ Avoids philosophy and psychology
- Disadvantages?

Introduction ... Cont'd

- ▶ AI follows all the four approach but tension existing between approaches which are centered around human and rationality.
- ▶ Human centered approach must be empirical science, involving **hypothesis** and **experimental confirmation**.
- ▶ A rationalist approach involves a combination of **mathematics** and **engineering**
- ▶ This course focus on the study of a rational agent that think and act rationally.

AI prehistory

- ▶ Philosophy Logic, methods of reasoning, mind as physical system foundations of learning, language, rationality
 - ▶ Mathematics Formal representation and proof algorithms, computation, (un) decidability, (in) tractability, probability
 - ▶ Economics Decision theory
 - ▶ Neuroscience physical substrate for mental activity
 - ▶ Psychology phenomena of perception and motor control, experimental techniques
 - ▶ Computer engineering building fast computers
 - ▶ Control theory design systems that maximize an objective function over time
 - ▶ Linguistics knowledge representation, syntax, grammar
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History of AI

▶ **Warren McCulloch and Walter Pitts (1943)**

- ▶ 1st AI work: Boolean circuit model of the brain
- ▶ Drew on three sources
 1. Knowledge of the basic physiology and function of neurons in the brain
 2. The formal analysis of propositional logic due to Russell and Whitehead
 3. Turing's theory of computation
- They proposed a model of artificial neurons
- They showed any computable function could be computed by some network of connected neurons
- They also suggested that suitably defined networks could learn

History of AI

- ▶ **Claude Shannon (1950) and Alan Turing (1953)**
 - Write a chess program
- **Marvin Minsky and Dean Edmonds (1951)**
 - Built the 1st neural network computer

History of AI Cont..

- ▶ Newell and Simon develop a reasoning program called the **Logic Theorist** (LT) before Dartmouth workshop
- ▶ They then come up with the **General Problem Solver** (GPS)
- ▶ **GPS**, unlike **LT**, is designed to imitate human problem solving protocols and it is the 1st program to embody the “thinking humanly” approach of AI
- ▶ Herbert Gelernter, 1959 constructed the **Geometry Theorem Prover** (GTP)

History of AI Cont..

- ▶ Arthur Samuel, 1952 wrote a series of checker programs
 - ▶ It can learn, which disprove the idea that computer can only do what they are told to do
 - ▶ His program play better than the creator
- ▶ John Mcarty, 1958 at MIT
 - ▶ Define the 2nd old high level programming language, LISP which is the 1st and dominant AI programming language
 - ▶ Invent time sharing concept with his friend to avoid the problem of time as a resource.

History of AI Cont..

- ▶ 1958 is also marked the year that Marvin Minsky moved to MIT
- ▶ Minsky supervised a series of students who choose limited problems that appeared to require intelligence to solve (this problems known as **Micro world**)
- ▶ The most dominant micro world problem is the block world
- ▶ It consists of a set of solid blocks placed on top of the table

Summary on history of AI

- ▶ 1943 McCulloch & Pitts: Boolean circuit model of brain
- ▶ 1950 Turing's "Computing Machinery and Intelligence"
- ▶ 1956 Dartmouth meeting: "Artificial Intelligence" adopted
- ▶ 1952—69 Look, Ma, no hands!
- ▶ 1950s Early AI programs, including Samuel's checkers program, Newell & Simon's Logic Theorist, Gelernter's Geometry Engine
- ▶ 1965 Robinson's complete algorithm for logical reasoning
- ▶ 1966—73 AI discovers computational complexity
Neural network research almost disappears
- ▶ 1969—79 Early development of knowledge-based systems
- ▶ 1980-- AI becomes an industry
- ▶ 1986-- Neural networks return to popularity
- ▶ 1987-- AI becomes a science
- ▶ 1995-- The emergence of intelligent agents

AI state of the art

- ▶ What can AI do today?
- ▶ A concise answer is difficult because there are so many activities in so many subfields.
 - ▶ Robotic vehicles
 - ▶ Speech recognition:
 - ▶ Autonomous planning and scheduling
 - ▶ Game playing
 - ▶ Spam fighting
 - ▶ Logistics planning
 - ▶ Robotics
 - ▶ Machine Translation

Summary

- ▶ Different people approach AI with different goals in mind. Two important questions to ask are: Are you concerned with **thinking or behavior**? Do you want to model **humans or work from an ideal standard**?
- ▶ Intelligence is concerned mainly with rational action. Ideally, an intelligent agent takes the best possible action in a situation.
- ▶ We study the problem of building agents that are intelligent in this sense.
- ▶ The history of AI has had cycles of success, misplaced optimism, and resulting cutbacks in enthusiasm and funding. There have also been cycles of introducing new creative approaches and systematically refining the best ones.

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- ▶ AI has advanced more rapidly in the past decade because of greater use of the scientific method in experimenting with and comparing approaches.
 - ▶ Recent progress in understanding the theoretical basis for intelligence has gone hand in hand with improvements in the capabilities of real systems.
 - ▶ The subfields of AI have become more integrated, and AI has found common ground with other disciplines

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- ▶ Write the four ways in which AI is defined with scenario.
 - ▶ Discuss the challenges and Opportunities of AI? Give example for each.